

Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence

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August 26, 2025

Abstract

This paper examines changes in the labor market for occupations exposed to generative artificial intelligence using high-frequency administrative data from the largest payroll software provider in the United States. We present six facts that characterize these shifts. We find that since the widespread adoption of generative AI, early-career workers (ages 22-25) in the most AI-exposed occupations have experienced a 13 percent relative decline in employment even after controlling for firm-level shocks. In contrast, employment for workers in less exposed fields and more experienced workers in the same occupations has remained stable or continued to grow. We also find that adjustments occur primarily through employment rather than compensation. Furthermore, employment declines are concentrated in occupations where AI is more likely to *automate*, rather than *augment*, human labor. Our results are robust to alternative explanations, such as excluding technology-related firms and excluding occupations amenable to remote work. These six facts provide early, large-scale evidence consistent with the hypothesis that the AI revolution is beginning to have a significant and disproportionate impact on entry-level workers in the American labor market.

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§Thanks to Nick Bloom, Joshua Gans, David Autor, Daniel Rock, Fei-Fei Li, Frank Li, Christina Langer, Sarah Bana, Cody Cook, Chris Forman, Andrew Wang, Brad Ross, Omeed Maghzian, Basil Halperin, Jiaxin Pei, Phil Trammell, Eric Bergman and participants at the Stanford Digital Economy Lab workshop for helpful feedback. We are grateful to ADP for access to the data and the Stanford Digital Economy Lab for financial support. All errors are our own.

¶Latest version: <https://digitaleconomy.stanford.edu/publications/canaries-in-the-coal-mine/>

1 Introduction

The proliferation of generative artificial intelligence (AI) has sparked a global debate about its potential impact on the labor market. This discourse, across academia, public policy, business, and popular media, spans utopian predictions of enhanced productivity, dystopian fears of widespread job displacement, and skeptical views that AI will have minimal effects on employment or productivity. Historically, technologies have affected different tasks, occupations, and industries in different ways, replacing work in some, augmenting others, and transforming still others. These heterogeneous effects suggest that there may be “canaries in the coal mine” which are harbingers of more widespread effects of AI.

There have been rapid improvements in AI capabilities in several areas. For instance, according to the most recent AI Index Report, AI systems could solve just 4.4% of coding problems on SWE-Bench, a widely used benchmark for software engineering, in 2023, but performance increased to 71.7% in 2024 (Maslej et al., 2025).¹ AI has improved on other benchmarks as well including language understanding, subject knowledge, and reasoning. At the same time, AI systems are increasingly widely adopted. According to Hartley et al. (2025), LLM adoption at work among U.S. survey respondents above age 18 reached 46% by June/July 2025.²

Given the better capabilities and widespread adoption, a central concern, amplified in recent headlines, is whether AI is beginning to supplant human labor, particularly for younger, entry-level workers in highly exposed professions like software engineering and customer service.³

Despite the intensity of this debate, empirical evidence has struggled to keep pace with technological advancement, leaving many fundamental questions unanswered. This paper confronts this empirical gap by leveraging a large-scale, high-frequency administrative dataset from ADP, the largest payroll software provider in the United States. Our sample consists of monthly, individual-level payroll records through July 2025, encompassing millions of workers across tens of thousands of firms. This rich panel structure allows us to track employment dynamics with a high degree of

¹SWE-bench is designed to evaluate the performance of large language models (LLMs) on real-world software engineering tasks. It uses a collection of GitHub issues to assess an LLM’s ability to generate code that resolves those issues.

²Similarly, Bick et al. (2024) found that in late 2024, nearly 40% of the U.S. population age 18-64 reported using generative AI, with 23% of employed respondents saying they had used generative AI for work at least once in the previous week, and 9% every work day.

³Improved productivity of workers in an occupation could lead to either reduced or increased employment, depending on, among other things, how elastic demand is for the output of those workers.

granularity, providing a near real-time view of labor market adjustments. By linking this data to established measures of occupational AI exposure and other variables, we can quantify the realized employment changes since the widespread adoption of generative AI.

This paper systematically presents six key facts that emerge from the data, offering an assessment of how the AI revolution is reshaping the American workforce.

Our first key finding is that we uncover substantial declines in employment for early-career workers (ages 22-25) in occupations most exposed to AI, such as software developers and customer service representatives. In contrast, employment trends for more experienced workers in the same occupations, and workers of all ages in less-exposed occupations such as nursing aides, have remained stable or continued to grow.

Our second key fact is that overall employment continues to grow robustly, but employment growth for young workers in particular has been stagnant since late 2022. In jobs less exposed to AI young workers have experienced comparable employment growth to older workers. In contrast, workers aged 22 to 25 have experienced a 6% decline in employment from late 2022 to July 2025 in the most AI-exposed occupations, compared to a 6-9% increase for older workers. These results suggest that declining employment AI-exposed jobs is driving tepid overall employment growth for 22- to 25- year-olds as employment for older workers continues to grow.

Our third key fact is that not all uses of AI are associated with declines in employment. In particular, entry-level employment has declined in applications of AI that *automate* work, but not those that most *augment* it. We distinguish between automation and augmentation empirically using estimates of the extent to which observed queries to Claude, the LLM, substitute or complement for the tasks in that occupation. While we find employment declines for young workers in occupations where AI primarily automates work, we find employment growth in occupations in which AI use is most augmentative. These findings are consistent with automative uses of AI substituting for labor while augmentative uses do not.

Fourth, we find that employment declines for young, AI-exposed workers remain after conditioning on firm-time effects. One class of explanations for our patterns is that they may be driven by industry- or firm-level shocks such as interest rate changes that correlate with sorting patterns by age and measured AI exposure. We test for a class of such confounders by controlling for firm-time effects in an event study regression, absorbing aggregate firm shocks that impact all workers at a